

Semester	T.E. Semester V– BIOM
Subject	Biomedical Digital Signal Processing
Subject Professor In-charge	Prof. Geetha Narayanan
Student Name	Rehan Raza Shaikh
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Grade and Subject Teacher's Signature	

Week 3 Submission

Project Title	Phonocardiogram (PCG)
Weekly Objective	Z-transform, Poles / Zero , Frequency , Filter
Key concepts learned	Z transform Signal in pcg Poles of pcg
Work Completed	Poles 0-100 sample Z- transform
Code	<pre>!pip install wfdb import wfdb # Download the MIT-BIH Arrhythmia Database (mitdb) which contains record '100'. # This ensures the necessary .dat, .hea, and .atr files are available locally. # The files will be downloaded to a default location, typically the current working directory # or a subdirectory named after the database. wfdb.dl_database('mitdb', dl_dir='.') # Download to the current directory</pre>

```
!pip install wfdb

import wfdb

import numpy as np

import matplotlib.pyplot as plt

from symprecord = wfdb.rdrecord('100')

print("Sampling Frequency:", record.fs)

print("Signal Shape:", record.p_signal.shape)

# Read first channel

pcg = record.p_signal[:, 0]

print("Total Samples:", len(pcg))

y import symbols, pprint

plt.figure(figsize=(15,4))

plt.plot(pcg)

plt.title("PCG Signal")

plt.xlabel("Sample")

plt.ylabel("Amplitude")

plt.grid(True)

plt.show()

x = pcg[:100]

plt.figure(figsize=(10,4))

plt.stem(range(100), x)

plt.title("First 100 Samples of PCG")
```

```
plt.xlabel("n")

plt.ylabel("Amplitude")

plt.grid(True)

plt.show()

z = symbols('z')

Xz = sum(float(x[n]) * z**(-n) for n in range(len(x)))

print("Z-Transform:\n")

pprint(Xz)

coeff = x[::-1]

zeros = np.roots(coeff)

print("Number of zeros:", len(zeros))

x = pcg[:100]

import numpy as np

# Polynomial coefficients (highest power first)

coeff = x[::-1]

# Compute zeros

zeros = np.roots(coeff)

print("Zeros:")

print(zeros)

num_poles = len(x) - 1

print("Number of poles:", num_poles)

print("All poles are located at z = 0")
```

```
import matplotlib.pyplot as plt

plt.figure(figsize=(7,7))

# Plot zeros
plt.scatter(np.real(zeros), np.imag(zeros),
            marker='o', s=60, label='Zeros')

# Plot poles at origin
plt.scatter(0, 0, marker='x', s=150, linewidths=2, label='99 Poles')

# Unit circle
theta = np.linspace(0, 2*np.pi, 500)
plt.plot(np.cos(theta), np.sin(theta), '--')

plt.axhline(0, color='black')
plt.axvline(0, color='black')

plt.xlabel("Real")
plt.ylabel("Imaginary")
plt.title("Pole-Zero Plot of PCG Signal")
plt.grid(True)
plt.axis('equal')
plt.legend()
plt.show()

import numpy as np
```

```
import matplotlib.pyplot as plt

# Sampling frequency from the header
fs = record.fs

# FFT
N = len(pcg)
fft_values = np.fft.fft(pcg)
freq = np.fft.fftfreq(N, d=1/fs)

# Positive frequencies only
positive = freq >= 0

plt.figure(figsize=(12,5))
plt.plot(freq[positive], np.abs(fft_values)[positive])
plt.title("Frequency Spectrum of PCG Signal")
plt.xlabel("Frequency (Hz)")
plt.ylabel("Magnitude")
plt.grid(True)
plt.show()

x = pcg[:100]

N = len(x)
fft_values = np.fft.fft(x)
freq = np.fft.fftfreq(N, d=1/fs)
```

```
positive = freq >= 0
```

```
plt.figure(figsize=(10,5))
```

```
plt.stem(freq[positive], np.abs(fft_values)[positive])
```

```
plt.title("Frequency Spectrum of First 100 PCG Samples")
```

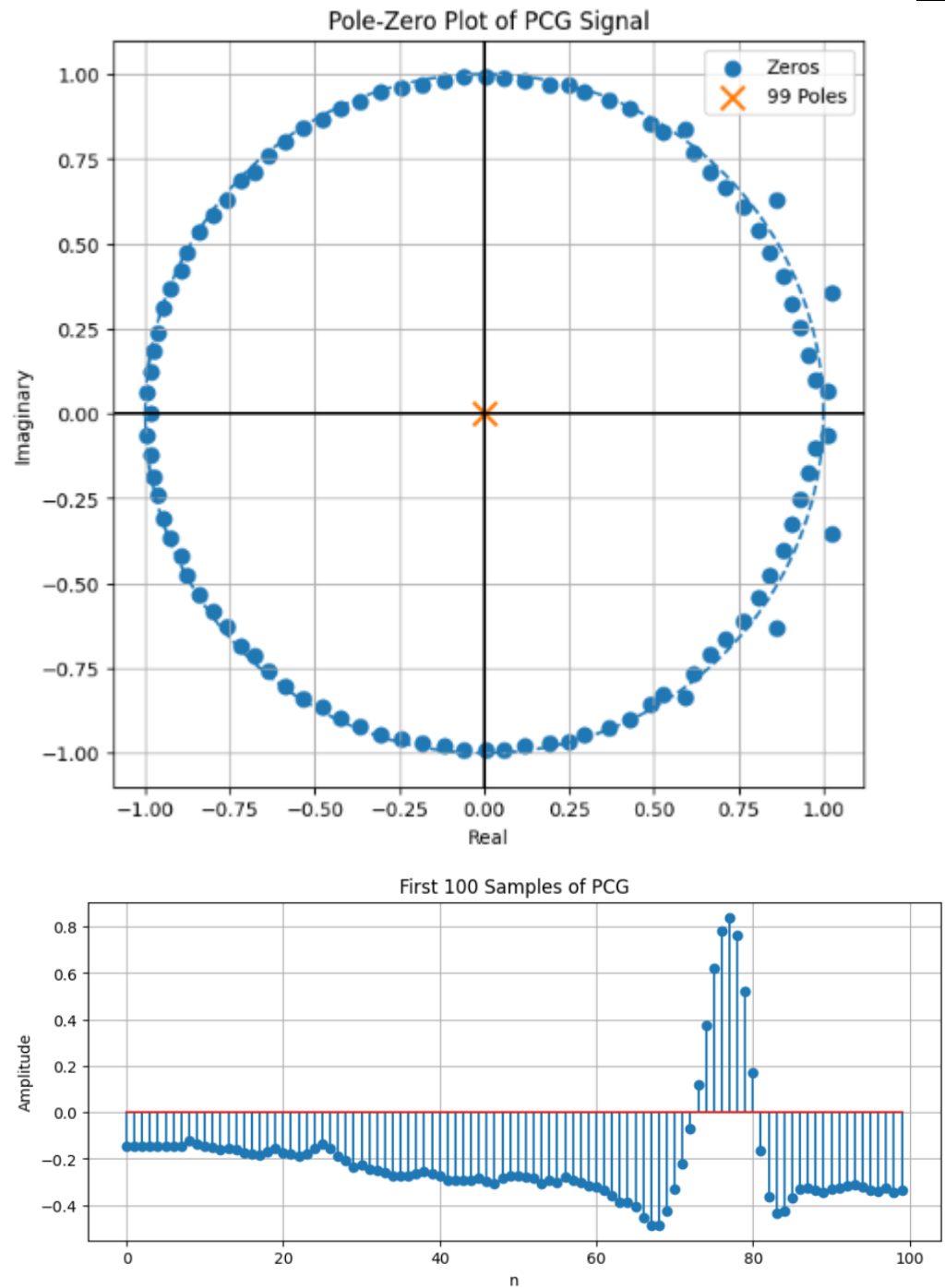
```
plt.xlabel("Frequency (Hz)")
```

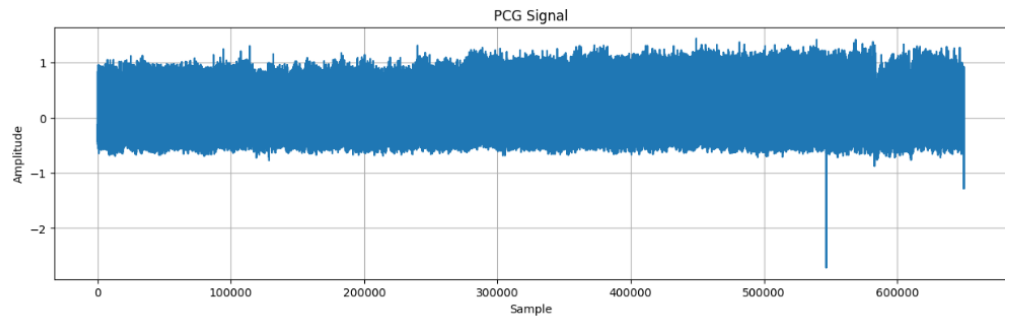
```
plt.ylabel("Magnitude")
```

```
plt.grid(True)
```

```
plt.show()
```

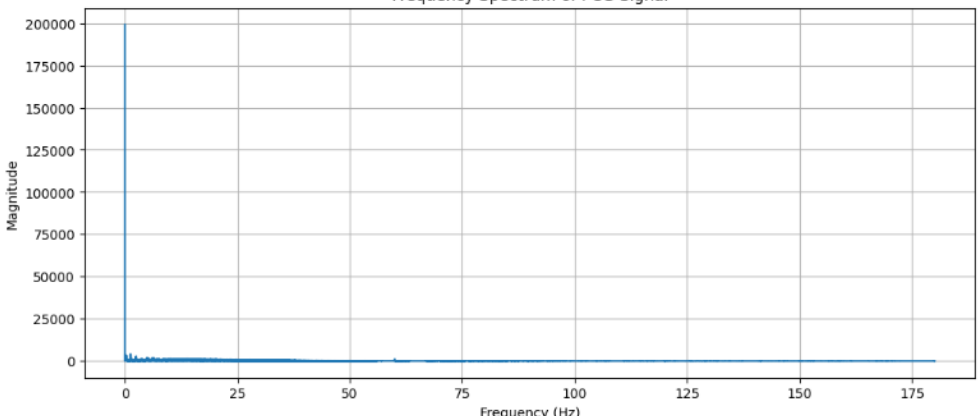
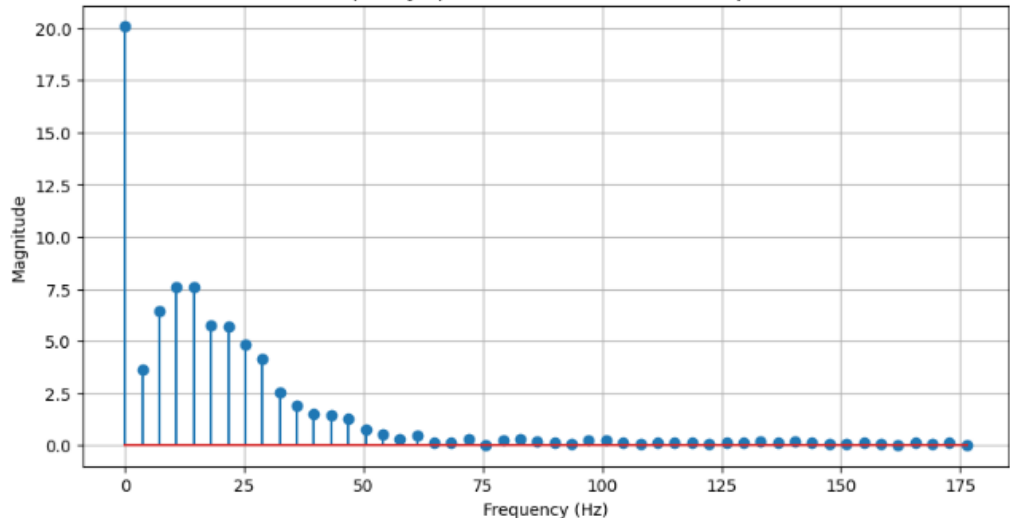
Evidence of
Progress/Results





... Z-Transform:

$$\begin{aligned}
 & -0.145 - \frac{0.145}{z} - \frac{0.145}{z^2} - \frac{0.145}{z^3} - \frac{0.145}{z^4} - \frac{0.145}{z^5} - \frac{0.145}{z^6} - \frac{0.145}{z^7} - \frac{0.12}{z^8} - \frac{0.135}{z^9} \quad \hookrightarrow \\
 & \hookrightarrow \frac{0.145}{z^{10}} - \frac{0.15}{z^{11}} - \frac{0.16}{z^{12}} - \frac{0.155}{z^{13}} - \frac{0.16}{z^{14}} - \frac{0.175}{z^{15}} - \frac{0.18}{z^{16}} - \frac{0.185}{z^{17}} - \frac{0.17}{z^{18}} - \frac{0.155}{z^{19}} \quad \hookrightarrow \\
 & \hookrightarrow \frac{0.175}{z^{20}} - \frac{0.18}{z^{21}} - \frac{0.19}{z^{22}} - \frac{0.18}{z^{23}} - \frac{0.155}{z^{24}} - \frac{0.135}{z^{25}} - \frac{0.155}{z^{26}} - \frac{0.19}{z^{27}} - \frac{0.205}{z^{28}} - \frac{0.235}{z^{29}} \quad \hookrightarrow \\
 & \hookrightarrow \frac{0.225}{z^{30}} - \frac{0.245}{z^{31}} - \frac{0.25}{z^{32}} - \frac{0.26}{z^{33}} - \frac{0.275}{z^{34}} - \frac{0.275}{z^{35}} - \frac{0.275}{z^{36}} - \frac{0.265}{z^{37}} - \frac{0.255}{z^{38}} - \frac{0.265}{z^{39}} \quad \hookrightarrow \\
 & \hookrightarrow \frac{0.275}{z^{40}} - \frac{0.29}{z^{41}} - \frac{0.29}{z^{42}} - \frac{0.29}{z^{43}} - \frac{0.29}{z^{44}} - \frac{0.285}{z^{45}} - \frac{0.295}{z^{46}} - \frac{0.305}{z^{47}} - \frac{0.285}{z^{48}} - \frac{0.275}{z^{49}} \quad \hookrightarrow \\
 & \hookrightarrow \frac{0.275}{z^{50}} - \frac{0.28}{z^{51}} - \frac{0.285}{z^{52}} - \frac{0.305}{z^{53}} - \frac{0.29}{z^{54}} - \frac{0.3}{z^{55}} - \frac{0.28}{z^{56}} - \frac{0.29}{z^{57}} - \frac{0.3}{z^{58}} - \frac{0.315}{z^{59}} - \frac{0}{z^{60}} \quad \hookrightarrow \\
 & \hookrightarrow \frac{.32}{z^{61}} - \frac{0.335}{z^{62}} - \frac{0.36}{z^{63}} - \frac{0.385}{z^{64}} - \frac{0.385}{z^{65}} - \frac{0.405}{z^{66}} - \frac{0.455}{z^{67}} - \frac{0.485}{z^{68}} - \frac{0.485}{z^{69}} - \frac{0.425}{z^{70}} \quad \hookrightarrow \\
 & \hookrightarrow \frac{0.33}{z^{71}} - \frac{0.22}{z^{72}} - \frac{0.07}{z^{73}} + \frac{0.12}{z^{74}} + \frac{0.375}{z^{75}} + \frac{0.62}{z^{76}} + \frac{0.78}{z^{77}} + \frac{0.84}{z^{78}} + \frac{0.765}{z^{79}} + \frac{0.52}{z^{80}} + \frac{0.1}{z^{81}} \quad \hookrightarrow \\
 & \hookrightarrow \frac{0.165}{z^{82}} - \frac{0.365}{z^{83}} - \frac{0.435}{z^{84}} - \frac{0.425}{z^{85}} - \frac{0.37}{z^{86}} - \frac{0.33}{z^{87}} - \frac{0.325}{z^{88}} - \frac{0.335}{z^{89}} - \frac{0.345}{z^{90}} - \frac{0}{z^{91}} \quad \hookrightarrow \\
 & \hookrightarrow \frac{0.325}{z^{92}} - \frac{0.315}{z^{93}} - \frac{0.31}{z^{94}} - \frac{0.32}{z^{95}} - \frac{0.335}{z^{96}} - \frac{0.34}{z^{97}} - \frac{0.325}{z^{98}} - \frac{0.345}{z^{99}} - \frac{0.335}{z^{100}} \quad \hookrightarrow
 \end{aligned}$$

	Frequency Spectrum of PCG Signal  Frequency Spectrum of First 100 PCG Samples 
Challenges & Solutions	We face error continuously in Z transform , Gemini code also give Error then I used chat gpt and gemini both and give proper command then we get our Z transform , poles and frequency signal,graphs
Next Week Plan	Frequency Filtering & week 4 concepts
Declaraion	I hereby declare that for the assignment / academic activity submitted by me, I confirm ONE of the following statements by ticking (☐) the appropriate box: ☐ (1) I have prepared the assignment / academic activity entirely by myself. ☐ (2) I have completed the assignment / academic activity with academic guidance

or discussion from seniors / friends / peers, while ensuring the work is my own.

☐ (3) I have used AI-based tools only as an aid and have appropriately acknowledged or cited their use as per the ethical guidelines of the Institute.

☐ (4) I have used AI-based tools and directly copy-pasted content for the assignment / academic activity.

☐ (5) I have copied the assignment / academic activity from peers / friends.

I understand that selecting options (4) or (5) indicates unethical academic practice and may attract academic penalties, including rejection of the assignment or disciplinary action, as per the rules of Vidyalkar Institute of Technology, Mumbai. I submit this declaration truthfully and accept full responsibility for the same.

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Signature: 

Date: 29/07/2026